

MA.6.GR.1.3

Benchmark

MA.6.GR.1.3 Solve mathematical and real-world problems by plotting points on a coordinate plane, including finding the perimeter or area of a rectangle.

Benchmark Clarifications:

Clarification 1: Instruction includes finding distances between points, computing dimensions of a rectangle or determining a fourth vertex of a rectangle.

Clarification 2: Problems involving rectangles are limited to cases where the sides are parallel to the axes.

Connecting Benchmarks/Horizontal Alignment

- MA.6.NSO.1.1, MA.6.NSO.1.2, MA.6.NSO.1.3, MA.6.NSO.1.4
- MA.6.NSO.4.1
- MA.6.AR.2.4

Terms from the K-12 Glossary

- Area
- Axes (of a graph)
- Coordinate
- Coordinate Plane
- Origin
- Quadrant

Vertical Alignment

Previous Benchmarks

- MA.5.GR.2.1

Next Benchmarks

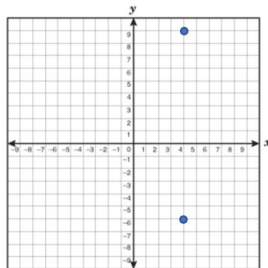
- MA.7.GR.1.1

Purpose and Instructional Strategies

In grade 5, students found the perimeter and area of rectangles using models and formulas. In grade 6, students plot the points on a coordinate plane and determine the perimeter or area. Students may also need to find the missing coordinate before determining the perimeter or area (*MTR.1.1, MTR.2.1*). In grade 7, students will extend their knowledge to find the areas of other quadrilaterals.

- Students will be able to find the area and perimeter of rectangles only if their side lengths are parallel to the axes (*MTR.6.1*).
- Instruction connects student understanding of MA.6.NSO.1 to the coordinate plane. Strategies include using absolute value to find the distance between two ordered pairs. Even though in grade 6, students are adding and subtracting integers, this benchmark focuses on using absolute value to explore addition and subtraction of rational numbers.

- For example, the points $(4, 9)$ and $(4, -6)$ are plotted on a coordinate grid.



Students can use the absolute value of the first y -coordinate, 9, and the second y -coordinate, -6 , and add these numbers together to determine the distance is 15.

- Instruction includes opportunities to find partial areas and partial perimeters in real-world context. When discussing area and perimeter, allow flexibility for students to use base and height or to use length and width.
 - For example, Nathaniel is building a garden in his backyard. His uncle has made a map of the backyard with a grid on it to help them plan out where the garden should go; the length of each box on the grid is equivalent to one meter. Nathaniel's garden has corners at $(0, 9)$, $(8, 9)$, $(0, 2)$ and $(8, 2)$. The y -axis represents the fence in his backyard. Find the perimeter to determine the amount of wood needed to create a barrier for the garden.

Common Misconceptions or Errors

- Students may switch the location of the x -coordinate and the y -coordinate in an ordered pair.
- Students may confuse perimeter (distance around a figure) with area (the total measure of the inside region of a closed two-dimensional figure).

Strategies to Support Tiered Instruction

- Instruction includes using two notecards and covering empty portions of the coordinate plane to focus attention on the space between two provided points. Students can then count the number of spaces between the two points, paying attention to scaling.
- If points are not already placed on a coordinate plane, students may plot the points to create a visual representation of the distances created.
- Instruction includes creating connections back to finding distance on a number line in order to determine the perimeter or area of a rectangle.
 - For example, the teacher can model finding the perimeter by laying a piece of tracing paper on top of the provided coordinate plane, trace the points, and draw a number line through the two points of one side of the rectangle, paying close attention to the scaling. Once the number line is down, remove the tracing paper and find the distance between the two points on the number line. Repeat this for all sides and then add the distances together to determine the perimeter.
 - For example, the teacher can model finding the area by laying a piece of tracing paper on top of the provided coordinate plane, trace the points, and draw a number line through the two points of the length (or base) of the rectangle, paying close attention to the scaling. Once the number line is down, remove the tracing paper and find the distance between the two points on the number line. Repeat

this for the width (or height) and then multiply the length (or base) and width (or height) together to determine the area.

- Instruction includes the use of geometric software to help build upon the concepts of area and perimeter on a coordinate plane.

Instructional Tasks

Instructional Task 1 (MTR.5.1, MTR.6.1)

A square has a perimeter of 36 units. One vertex of the square is located at $(3, 5)$ on the coordinate grid.

Part A. What could be the x - and y -coordinates of another vertex of the square?

Part B. What is the area of the square?

Instructional Task 2 (MTR.3.1, MTR.5.1)

Sandy wants to find the area of a rectangular garden where one side is a side of her house.

She graphed the garden on a coordinate plane so that three of the vertices are at:

$(-3, -2)$, $(4, -2)$ and $(4, 4)$.

Part A. Find the coordinates of the fourth vertex so that the garden is a rectangle.

Part B. Find the area of the garden, showing your work.

Part C. If Sandy wants to enclose the garden, what is the length of fencing needed?

Instructional Items

Instructional Item 1

The corners of a rectangular swimming pool are located at $(-4, -3)$, $(-4, -8)$, $(6, -3)$ and $(6, -8)$. What is the perimeter of the swimming pool?

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*